Prevalence of Transfusion-Transmissible Infections in Donors to an Ethiopian Blood Bank Between 2009 and 2013 and Donation Factors That Would Improve the Safety of the Blood Supply in Underdeveloped Countries

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ABSTRACT

Objective: To determine the percentages of blood donors at an Ethiopian blood bank who tested positive for 4 transfusion-transmissible infections (TTIs) and to compare TTI infection levels among individuals of different sociodemographic characteristics.

Method: We reviewed 5 years of data (2009 to 2013) from registration records of blood donors at the blood bank of Yirgalem Hospital in Hawassa, Ethiopia.

Result: Of the 6367 donors, 447 (7.0%) tested positive for 1 of the TTIs for which infection necessitates discarding the infected blood unit. The prevalence of HIV (human immunodeficiency virus), HBV (hepatitis B virus), HCV (hepatitis C virus), and syphilis were 1.6%, 4.8%, 0.6%, and 0.5%, respectively. The prevalence of HIV was significantly increased among donors who were replacing blood for family members compared with volunteer donors (adjusted odds ratio [AOR], 0.34; 95% CI, .15 to .77), and employed donors compared with students (2.48; 1.22 to 5.02). HCV prevalence was significantly increased with replacement donors compared with volunteers. The prevalence of the 4 studied TTIs in volunteers' donations are lower than that in family replacement donation. The discard rate of blood from volunteers was 1.1%, versus 5.9% from replacement donors.

Conclusion: The prevalence of TTIs is significant, and blood collected from volunteer donors is associated with lower risk of TTIs. Therefore, agencies must ensure a safe, adequate blood supply, designating unremunerated volunteers as a preferred donor group. Moreover, more sensitive and specific screening methods and a regular supply of materials and reagents must be provided to improve the quality standard of blood-bank laboratories.

Keywords: TTIs, volunteer donors, prevalence, blood component, malaria, discard rate

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Although blood transfusion can be a life saving type of therapy for medical and surgical patients, unsafe transfusion practices can also put millions of people at risk of transfusion transmitted infections (TTIs).[1–3] Infectious diseases, such as human immunodeficiency virus (HIV), syphilis, hepatitis, and malaria are leading concerns and priorities for prevention and control in the blood transfusion strategies and programs of the World Health Organization (WHO), particularly in Sub-Saharan African countries such as Ethiopia.[4] Although infectious diseases have many transmission routes, blood donation can be significant avenues for transmission of these diseases to healthy individuals.[3.]

Most donated blood in Ethiopia is not screened adequately for TTIs due to the lack of a consistent supply of screening reagents and materials. This is a major challenge to the transfusion services in Ethiopia and throughout the underdeveloped world.[4,5] As a result, use of unscreened blood puts the patient at risk of acquiring many TTIs. Such patients can become potential sources of infection to the

Abbreviations

TTIs, transfusion transmitted infections; HIV, human immunodeficiency virus; WHO, World Health Organization; YAH, Yirgalem Hospital; IRB, Institutional Review Board; HBV, hepatitis B virus; HCV, hepatitis C virus; HBsAg, hepatitis B surface antigen; ELISA, enzyme-linked immunosorbent assay; RPR, rapid plasma reagin; NGOs, nongovernmental organizations; AOR, adjusted odds ratio

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community.[2] Consequently, disease burden can impact the entire country and can trigger additional costs for diagnosis and treatment of diseases.[3]

To ensure the safety and adequacy of the national blood supply in Ethiopia, development of a national blood policy, mobilization and recruitment of low-risk donors, strengthening regional blood banks, and testing 100% of donated blood units for TTIs are essential. Providing adequate training for prescribers and users of blood and blood products are also critical to improve the clinical use of blood.[4]

The aims of this study were to determine the prevalence of and trends in TTIs, which will further help assess the safety of blood transfusion as therapy at Yirgalem Hospital (YAH) in Hawassa, Ethiopia. This study also emphasizes material and reagent supply for the diagnosis of TTIs in the hospital blood bank laboratory.

Materials and Methods

I reviewed 5 years of data (2009 to 2013) from registration records of blood donors at the Yirgalem Hospital blood bank. A total of 6337 donors with complete survey information were included in the study. I prepared a data input form to submit TTIs results, gathering information that included sociodemographic and blood type differences. I sought ethical clearance to undertake this study from the Institutional Review Board (IRB) of Hawassa University College of Medicine and Health Sciences, Hawassa, Ethiopia; permission was obtained from Yirgalem Hospital administrators to collect the data. The study assessed the prevalence of TTIs (syphilis, hepatitis B virus [HBV], hepatitis C virus [HCV], and HIV) and determined factors and donor groups associated with provision of blood screened as testing negative for infectious diseases. Donors at the blood bank center of YAH would be eligible for donation after screening criteria established by the Ethiopian Red Cross Society.

The YAH blood bank laboratory was used as a zonal center; also, the Ethiopian Red Cross Society provided materials and reagents during the study period. I screened for HIV 1 and HIV 2 using the Vironostika Uni-form II Ag/Ab 4th-generation HIV test (bioMérieux SA) and for hepatitis B surface antigen (HBsAg) using the DIALAB test (DIALAB GmbH). I performed enzyme-linked immunosorbent assay (ELISA), (specificity in blood donors, 99.87%); also, HCV antibody was screened for using DIALAB testing and ELISA (sensitivity, 99.79%).

Syphilis was screened for using the rapid plasma reagin (RPR)–carbon technique and linear chemicals; the diagnostic sensitivity was 86% (primary syphilis) and 100% (secondary syphilis), respectively. I determined the ABO blood group using the monoclonal blood grouping antisera anti A and anti B; Rh(D) testing was performed using anti D antisera. All procedures were carried out according to the manufacturer instructions. Due to the large amount of data, I performed double verification of the accuracy of data entry. Data were analyzed using SPSS software, version 19 (SPSS, Inc). I presented percentages; the association between variables was calculated using χ^2 and binary logistic regression. I considered associations to be statistically significant when their *P* value was less than .05.

Results

Study Population

In this survey, I examined a total of 6337 donor records for TTIs and sociodemographic factors. Among those donors, 86.5% were male and 46.3% were aged 21 years to 30 years. Students and farmers 79.7% of the blood donor pool. Most (79.1%) of donations were replacement (family) type. The O-type blood group was the most common (42.6%), and specimens from 340 (5.4%) tested rhesus negative (**Table 1**).

I assumed that *students* are full-time students who do not have an occupation; *employed* individuals are study participants who have at least a certificate of education and who work in governmental (public) or nongovernmental organizations (NGOs). The category *other* refers to a small group of study participants that includes drivers, housewives, military personnel, self-employed persons, and members of the clergy.[2] *Family/replacement blood donors* indicates donors who give blood when it is required by a member of the family or community of the patient to replace units used. This may involve a hidden paid donation system in which the donor is paid by the family of the patient.

Volunteer vs Replacement Donors

Overall, among the individuals in the donation pool, 79.1% were replacement/family donations and 20.9% were

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Table 1. Demographic Characteristics of Blood Donors[a]

Characteristic		No (%)[b]
Age, y	15-20	1789 (28.2%)
	21-30	2931 (46.3%)
	31-40	1098 (17.3)
	41-75	519 (8.2)
Sex	Female	835 (13.2)
	Male	5471 (86.8%)
Occupation	Student	2424 (39.9%)
	Farmer	2415 (39.8%)
	Employed[c]	518 (8.5%)
	Trader	533 (8.8%)
	Other[d]	183 (3.0%)
Donor Type	Replacement	5010 (79.1%)
	Volunteer	1327 (20.9%)
ABO	O	2683 (42.6%)
	A	1746 (27.7%)
	B	1481 (23.5%)
	AB	382 (6.1%)
Rhesus test result	Positive	5949 (94.6%)
	Negative	340 (5.4%)

[a] *n* = 6337.
[b] Percentages may not total 100 because of rounding.
[c] Individuals who work in governmental (public) or nongovernmental organizations.
[d] Occupations include drivers, housewives, military personnel, self-employed persons, and members of the clergy.

volunteer donors. As indicated in **Figure 1**, donation of blood at the YAH blood bank laboratory steadily increased from 2009 to 2013. Specifically, most of the volunteer donors were in the 15-years-to-20-years age group, and 38% of replacement donors were in the 21-years-to-30-years age group. Males constituted most (75.0%) of the replacement donors, and females made up slightly less than males among volunteer donations. Of the volunteer donors, 20.0% were students, and most (39.4%) of the family (replacement) donations were made by farmers. Students participated in equal numbers ($n = 1212$) as volunteer and replacement donors. Farmers predominate in the replacement-donation data. Among 340 donors whose specimens tested Rhesus-negative, most (172 [50.6%]) were found to have the O blood type ($P = .008$).

Of the overall study population, 447 (7.1%) tested positive for a single TTI. The prevalence of HIV, HBV, HCV, and syphilis were 1.6%, 4.8%, 0.6%, and 0.5%, respectively. Among those 447 individuals, 12 donors had coinfection with 2 TTIs. The most common combinations were HIV-HBV (8[66.7%]), HIV-HCV (3[25.0%]), and HBV-HCV (1 [8.3%]).

The prevalence of HIV was higher among replacement (family)–type blood donors compared with volunteer donors

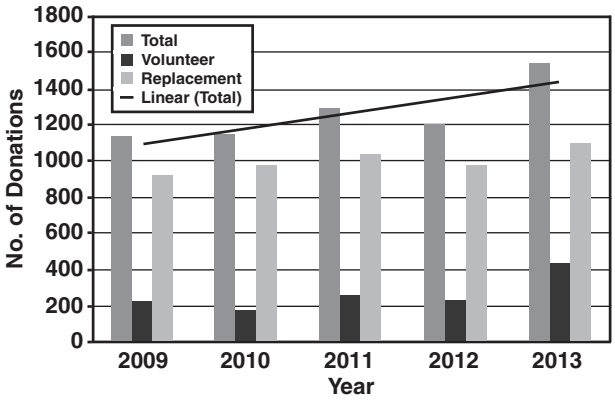


Figure 1

Total blood donation, volunteer vs replacement, in the years 2009 to 2013 at Yirgalem Hospital in Hawassa, Ethiopia.

($P = .001$) and among individuals who were employed compared with students ($P = .01$). HCV prevalence was significantly increased in the replacement donor type ($P = .03$) compared with volunteers (**Table 2**).

The prevalence of HBV, HCV, and syphilis among donors who tested HIV negative was 4.7%, 0.6%, and 0.5%, respectively; among donors who tested HIV positive, the prevalence rates were 8%, 2.9%, and 0%, respectively. Further, we observed a statistically significant association between HCV and HIV infection ($P = .002$). Also, the prevalence of the 4 TTIs examined herein has decreased over the years, as shown in **Figure 2**.

Discussion

Overall, 447 individuals (7.1%) tested seropositive for at least 1 of the screened pathogens, which accounts for a 1 in 14 discard rate. A report by Leena et al[1] indicated a 1.35% prevalence for all TTIs, and another report[2] demonstrated a 3.8% positive rate in Eritrea.

I observed a greater prevalence of TTIs in employed individuals (9.3%; $P = .02$), compared with individuals in the other occupation category groups and in replacement donors (7.5%; $P = .004$) compared with voluntary donors. Also, although the results were not statistically significant, TTIs were more common in the O blood type group, in males, and in individuals aged 41 years to 75 years.

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Table 2. Prevalence of TTIs Among Donors, by Donor Characteristics

Donor Characteristics	HIV (%)	HBV (%)	HCV (%)	Syphilis (%)	Overall TTIs[a]
Age, y					
15-20	1.6	4.6	0.4	0.1	115/1789 (6.4%)
21-30	1.4	5.2	0.7	0.6	217/2931 (7.4%)
31-40	1.7	4.4	0.7	0.7	74/1098 (6.7%)
41-75	2.5	4.4	0.6	1.6	41/519 (7.9%)
Sex					
Female	1.1	4.3	0.6	0	50/835 (6.0%)
Male	1.7	4.9	0.6	0.6	395/5471 (7.2%)
Occupation					
Student	1.1	4.7	0.5	0.2	156/2424 (6.4%)
Farmer	1.7	4.6	0.6	0.9	170/2245 (7.0%)
Employed[b]	3.1	5.8	1.2	0	48/518 (9.3%)
Trader	2.1	6.0	0.9	0	47/533 (8.8%)
Other[c]	0.5	2.2	0	0.7	6/183 (3.3%)
Donor Type					
Volunteer	0.6	4.5	0.2	0	70/1327 (5.3%)
Replacement	1.9	4.9	0.6	0.7	377/5010 (7.5%)
ABO Blood-Grouping System					
O	1.7	5.2	0.7	0.2	202/2683 (7.5%)
A	1.5	4.4	0.5	0.9	114/1746 (6.5%)
B	1.5	4.8	0.6	0.6	104/1481 (7.0%)
AB	1.0	4.5	0.5	0.6	23/382 (6.0%)
Rhesus test result					
Positive	1.6	4.8	0.6	0.5	420/5949 (7.1%)
Negative	1.5	5.0	0.6	0	23/382 (6.0%)

TTIs, transfusion-transmissible infections; HIV, human immunodeficiency virus; HBV, hepatitis B virus; HCV, hepatitis C virus
[a]*Percentages may not total 100 because of rounding.*
[b]*Individuals who work in governmental (public) or nongovernmental organizations.*
[c]*Occupations include drivers, housewives, military personnel, self-employed persons, and members of the clergy.*

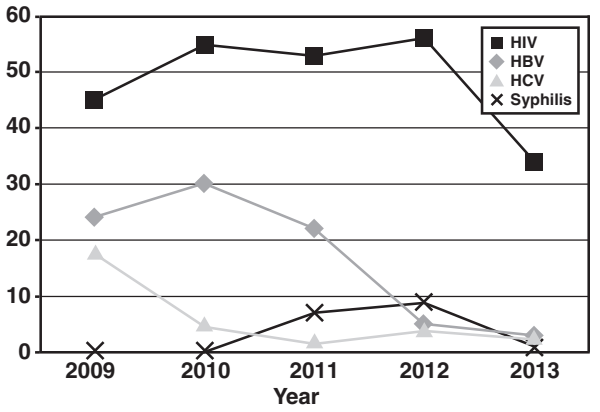


Figure 2

Percentage prevalence of transfusion-transmissible infections in the study period (2009 to 2013) at Yirgalem Hospital in Hawassa, Ethiopia.

As our results showed, 7.1% of the donated blood tested seropositive for at least 1 of the screened pathogens (yielding a 1 in 14 discard rate). This extremely high prevalence calls for stricter screening of donated blood and implementation of more stringent donor selection criteria. Among the 12 types of coinfection, the most common combination was HIV-HBV (67%),[8] which was reported as the second-most-prevalent coinfection type in Ethiopia (34%),[6] and in Nigeria (33.3%).[3] In this study, I discovered that HIV-HCV was the second-most-prevalent coinfection type, with a rate of 25%, similar to the rate of HIV-HCV in Nigeria (11.1%).[3] However, HIV-HCV was the third most prevalent coinfection type in Gondar, Ethiopia (8%).[6]

The prevalence of HIV infection among the blood donors at YAH is 1.6%. In this study, I found HIV to be more common in the age group of 41 years to 75 years (13/506 [2.6%]), in the donors who were employed (16/502 [3.2%]), and in male donors (93/5377 [1.7%]), although there were associations that were not statistically significant. The prevalence of HIV was increased among replacement family blood donors, compared with volunteer donors (adjusted odds ratio [AOR], 0.34; 95% CI, .15 to .77) and in employed individuals compared with students (2.48; 1.22-5.02; $P < .25$ in coefficient of reliability [COR] was considered for AOR).

In a family/replacement blood donor system, a donor gives blood when it is required by a member of the family or community of a patient. This process may involve a hidden paid-donation system in which the donor is paid by the family of the patient. Such donors are often economically deprived and are more likely associated with risky behavior and habits that expose them to HIV and other diseases.

Recent studies have indicated a lower prevalence of HIV among donors in various countries, namely, India (0.27%),[1] Pakistan (0.06%),[7] Eritrea (0.18%),[2] Nepal (0.12%),[8] and China (0.08%).[9] Comparable results have been reported from Cameroon (1.77%);[3] also, higher prevalence rates were reported from Nigeria (3.1%)[6] and Gondar, Ethiopia.[10] In this study, I report an HBV prevalence of 4.8%; this finding is comparable with that from Gondar (4.7%).[6] HBV was most commonly found in traders, (32/533 [6.0%], males (266/5467 [4.9%]), and in the groups of individuals aged 21 years to 30 years (151/2930 [5.2%]); these results, however, were not statistically significant. Studies in other countries

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reported lower prevalence of HBV than those from this study: from Eritrea, 2.58%;[2] from Nepal, 0.46%;[8] from India, 0.41%;[1] from China, 0.86%;[9] and from Pakistan, 2.68%.[7] A higher prevalence of HBV was observed in Nigeria, namely, 18.6%.[3]

The 0.6% HCV prevalence reported in this study was comparable with rates reported from China (0.57%);[2] Nepal (0.64%);[6] Gondar, Ethiopia (0.7%);[8] and Eritrea (0.57%).[9] Higher prevalence was reported from Nigeria (6%),[1] India (1.23%),[3] and Pakistan, 2.46%.[7] HCV prevalence is significantly increased within the replacement donor type ($P = .03$), compared with volunteers (COR, 0.28; 95% CI, .05-.87). Our study results reveal that HCV is more frequently found in employed individuals (6/512 [1.17%]), males (33/5436 [0.6%]), and replacement donors (36/5008 [0.7%]).

Despite the fact that I only had 3 years of data on the prevalence of syphilis infection, the percentage among the blood donors was 0.5%. All 15 cases of syphilis were found in males, those in the replacement donor group, and in donors who tested Rhesus positive. Also, syphilis was frequently found in farmers (9/1042 [0.9%]) and in the group of donors aged 21 years to 30 years (8/1403 [0.6%]). This finding is comparable with findings from reports from Eritrea (0.45%),[2] Nepal (0.48%),[7] China (0.47%),[8] and Pakistan (0.43%);[9] however, it is lower than the rates observed in Cameroon (8.1%),[3] Gondar (1.3%),[6] and Nigeria (1.1%).[10] Because this survey included only a 3-year data study, the results indicate that syphilis was not routinely tested for in the blood bank laboratory. This lack of screening is contrary to the Ethiopian blood transfusion service guideline that dictates that all blood must be screened for TTIs, including syphilis.[4] Also, more sensitive methods of diagnosing syphilis need to be used. Blood bank laboratories already should be screening for syphilis because there can be situations in which blood can be transfused immediately, without the refrigeration storage that has been believed colloquially to kill the bacteria *Treponema pallidum.*

The prevalence of HIV at my institution decreased from 2.4% in 2009 to 2.2% in 2011 and then decreased drastically to 0.5% in 2012 and 0.3% in 2013 ($P < .001$). Similarly, the prevalence of HCV decreased from 1.8% in 2009 to 0.5% in 2010 and to 0.2% in 2011; it then decreased to 0.4% in 2012 and to 0.3% in 2013 ($P < .001$; **Figure 2**). Similarly, in other surveys in Ethiopia, the prevalence of HIV among blood donors was shown to have drastically decreased, from 4.5%[11] to 3.8%,[6] and to the 1.6% that I report in this study. These findings may have resulted from the different intervention measures taken to prevent the spread of HIV during recent years in Ethiopia.

As a policy, voluntary nonremunerated blood donors should be recruited from low-risk population groups in all blood centers. There were 1143 (18%) donations in 2009, 1153 (18.0%) in 2010, 1295 (20.4%) in 2011, 1208 (19.1%) in 2012, and 1538 (24.3%) in 2013. Likewise, volunteer donation steadily increased through the years: in 2001, there were 220; in 2003, there were 263; and in 2005, there were 440. Contrary to these findings, the proportion of volunteer/replacement donation in Eritrea, on average, was 3.8%;[2] also, the percentage in Nigeria was 0.07%,[3] which is less than the 0.26% reported in my study in Ethiopia (0.26%). This finding indicates that, although the level of volunteer donation at Yirgalem Hospital appears to be slightly increasing, it is not yet satisfactory. Although efforts in recruiting volunteer, nonremunerated donors and standardization of the quality of service are improving, much work remains to be undertaken to develop the highest level of safety in blood transfusion service in the region.

Safer blood tends to come from volunteer, noncompensated donors. Research performed in several countries[4] has proven that blood collected from volunteer, nonremunerated donors is much safer than blood collected from replacement donors. Likewise, I report that the prevalence of TTI in volunteer donors is 5.3%, which is significantly less than the prevalence in replacement donors, namely, 7.5%. Specifically, the prevalence of HIV, HBV, HCV, and syphilis in volunteer' donation was 0.6%, 4.5%, 0.2%, and 0%, respectively, compared with 1.9%, 4.9%, 0.7%, and 0.5%, respectively, for replacement (family) donation.

The discard rate in volunteers was 1.1%, and the replacement rate was 5.9%, in total donations. Among the volunteer donors, students comprise the greatest proportion (91.4%). This finding indicates that students participate more frequently in volunteer-donation campaigns. Volunteer donation, as the results of this study have shown, is less risky for TTIs; therefore, volunteer donation must be encouraged to provide safer blood to meet clinical demand. Special programs must be organized to expand volunteer donation, which is yields a safer blood supply than other donation methods.

The Ethiopian national blood transfusion service guidelines emphasize a huge demand for safe blood and blood products in health institutions. Given these results, setting up and

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expanding a regional or national blood bank service would be the next step in enhancing the quality and efficiency of blood transfusion services in Ethiopia. For instance, blood banking in developed countries is regionally organized such that fresh stocks of blood are kept centrally and distributed daily to all hospitals with a blood bank. Modeling blood banking service in Ethiopia at a national or regional level, involving registered donors with regular donation sessions, would be a major but highly useful undertaking.

This survey identified additional issues that will contribute to the development of transfusion services in Ethiopia. For example, although malaria is endemic in Ethiopia, blood donations were not screened for malaria during the time of the study (most Ethiopian laboratories do not screen blood donors for malaria). Therefore, testing donors for malaria should be initiated and promoted.

In this study, donor blood was screened after donation, as is standard practice in most areas of Ethiopia. There is controversy regarding the timing of screening: should donors be screened before they donate blood, or should the donated blood be screened for TTIs only after the donation is complete? In countries with a high incidence of TTIs, screening donors before donation may stigmatize potential donors who test positive for 1 or more TTIs, especially HIV. This approach requires counseling by skilled professionals for infected donors. In the current blood bank service, there is no established systematic approach for counseling and managing donors who test positive for TTIs. Therefore, such concerns should be addressed by responsible agencies, such as the Ethiopian Blood Transfusion Service.

Conclusion and Recommendation

Overall, the prevalence of TTIs is high, at a time when blood therapy is widely used. Blood collected from volunteer donations is associated with less risk (prevalence) of TTIs. Therefore, responsible bodies must better organize their efforts to realize the availability of safer blood and the designation of nonrenumerated volunteers as a preferred donor group. Also, more sensitive and specific screening methods and a regular supply of materials and reagents must be required for the development of blood bank service in the region of Sub-Saharan Africa and other underdeveloped countries. **LM**

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